Literature Review

The Central Argument: A Gap Three Decades Wide

Existing quantitative frameworks for predicting post-acquisition synergy are fundamentally constrained by a shared architectural assumption: they treat the firm as an isolated data point, devoid of context, history, or structural position. Financial ratio models, machine-learning classifiers, and even transformer-based NLP pipelines all process firms as nodes without edges — objects detached from the industrial ecosystems that define their actual value. This literature review builds a sustained critical argument across four interconnected knowledge streams:

1. **Stream I — The M&A Paradox and the Failure of Valuation Theory**, which diagnoses *why* synergy prediction is difficult and demonstrates that the root cause is structural information loss, not computational insufficiency.
2. **Stream II — The Tabular Paradigm**, covering the full history of quantitative M&A models from logistic regression through gradient boosting and early deep learning, demonstrating that each generation inherits the same fatal independence assumption — and establishing the precise ceiling beyond which single-modality tabular models cannot advance.
3. **Stream III — The Semantic Turn**, tracing the evolution of financial NLP from bag-of-words lexicons to transformer architectures, critically demonstrating that despite increasing sophistication, the field systematically misdirects powerful tools at deal *occurrence* rather than deal *outcome*, and that even FinBERT implementations fall into a "Tabular Trap" the moment their embeddings are flattened into scalar vectors.
4. **Stream IV — The Topological Turn**, engaging with the supply chain finance, corporate networks, and graph neural network literatures to establish that network topology carries *irreducible* economic signal about synergy potential — signal that is mathematically unrecoverable by any non-graph model regardless of its parameter count.

The review concludes that these four streams converge on a single logical exit strategy: a heterogeneous graph architecture that simultaneously fuses financial fundamentals (Block A), domain-specific textual embeddings (Block B), and supply-chain/competitor graph topology (Block C) within a unified predictive framework — a configuration that, as of 2025, no published study has deployed for post-merger synergy *outcome* prediction. Every section below is therefore not a neutral survey but a *motivated critique*: prior work is evaluated for what it structurally *cannot* achieve, and each limitation is mapped directly to one of the three hypotheses tested in this dissertation.

Stream I — The M&A Paradox and the Failure of Valuation Theory

The Empirical Record: Stationarity of Failure

Mergers and acquisitions remain the primary mechanism for global capital reallocation, yet their empirical track record constitutes one of the most durable anomalies in corporate finance. Martynova and Renneboog [1] documented that between 70% and 90% of acquisitions fail to generate value for acquirer shareholders across a full century of global takeover activity — a finding replicated across multiple independent meta-analyses [2]. Critically, this failure rate has exhibited *statistical stationarity* across distinct economic regimes — from the conglomerate wave of the 1960s through the dot-com era and post-financial-crisis consolidations — implying that the root cause is structural rather than cyclical [1].

Bradley, Desai, and Kim [3] established that while target shareholders captured average announcement-date abnormal returns of approximately 30%, acquirer shareholders experienced systematic losses, yielding combined synergy gains averaging only 7.4% of combined pre-deal firm value. This gap between the value created for targets and the value transferred (or destroyed) for acquirers is the precise economic surplus that a reliable predictive model should identify *ex ante*. The persistence of this destruction in the face of progressively more sophisticated computational due diligence tools motivates the epistemological question this study addresses: what information, structurally absent from conventional models, would close this predictive gap?

Behavioural Distortions: Hubris as Label Noise

The first explanatory layer is behavioural. Roll's Hubris Hypothesis [4] argues that acquiring managers systematically overestimate their ability to extract value, treating the target's efficient market price as an underestimate rather than an unbiased signal. This creates the "Synergy Trap" identified by Sirower [5], wherein the premium paid mathematically requires performance improvements that are unattainable in the majority of cases.

From a machine learning perspective, hubris introduces systematic *label noise* into training datasets: if acquisition premiums partly reflect executive overconfidence rather than genuine synergy potential [4], models trained on transaction values are fitting a signal contaminated with behavioural error. This contamination is *irreducible* — no amount of regularisation or hyperparameter tuning can remove bias that is encoded into the target variable itself. This motivates the use of Cumulative Abnormal Return (CAR) as the prediction target rather than deal premia or accounting synergies, since CAR represents the market's *ex post* revision of expectations and is less susceptible to managerial narrative.

Market Timing and the Stock-Driven Wave

A complementary source of label contamination is identified by Shleifer and Vishny [6], whose stock-market-driven acquisitions model demonstrates that overvalued acquirers rationally exploit inflated equity as acquisition currency during market peaks. Under this framework, a substantial proportion of M&A waves are motivated by relative misvaluation rather than operational complementarity — the observed CAR reflects market correction dynamics rather than synergy realisation. Grossman and Stiglitz [7] established the theoretical bound here: in a world of costly information acquisition, markets cannot be fully informationally efficient, and persistent information asymmetries around deal announcements are a natural equilibrium. This "bounded efficiency" framing justifies the premise that pre-announcement features can contain predictive signal about CAR direction — information costs prevent instantaneous arbitrage of the signal this study attempts to extract.

The Epistemological Barrier: Structural Blindness

Even setting aside behavioural distortions, the verification mechanism is structurally deficient. Angwin [8] documented that conventional due diligence operates in organisational silos — financial, legal, and operational teams independently assess their respective domains without modelling interaction effects. This orthogonal approach misses multiplicative risks: strong cash flows combined with a fragile single-source supplier dependency produce a combined exposure that sums to near-zero value; linear due diligence passes both in isolation and misses the conjunction.

Akerlof's [9] "Market for Lemons" framework provides the theoretical grounding for this structural blindness: mandated disclosures (10-K filings) are insufficient to eliminate information

asymmetry because targets can strategically obscure ecosystem vulnerabilities within the noise of standard reporting formats [10]. The implication is precise: reliable synergy estimation requires information sources that cannot be strategically manipulated — specifically, the verifiable topology of the firm's external network of suppliers, customers, and competitors. This forms the primary theoretical motivation for Block C in this study's architecture.

Stream I gap: The M&A failure rate is predictable from structural information that is systematically absent from existing valuation models. The following three streams diagnose exactly which information each prior paradigm discards — and why.

Stream II — The Tabular Paradigm: An Asymptotic Ceiling

The Econometric Era: Linearity Bias and Its Persistence

The first generation of quantitative M&A predictors — Palepu [11], Barnes [12] — employed logistic regression and multiple discriminant analysis on financial ratios. These models operate under a "Linearity Bias": the assumption that financial metrics bear monotonic relationships to synergy potential. Barnes [12] demonstrated empirically that this assumption fails for leverage: moderate debt signals fiscal discipline; high debt signals distress. This creates a non-monotonic, context-dependent relationship that linear discriminants fundamentally cannot represent.

More critically, Palepu's [11] own out-of-sample validation found that acquisition targets could not be reliably predicted beyond chance — a result that exposed the information deficit in ratio-based models rather than any implementation flaw. The pseudo-R^2 ceiling below 0.10 documented consistently across econometric studies [13] is not a calibration failure; it is mathematical evidence that financial ratios do not contain sufficient information to reconstruct synergy outcomes. This ceiling is the first benchmark any successor model must demonstrably surpass.

The Machine Learning Escalation: Why XGBoost Hits a Wall

The emergence of ensemble methods promised to resolve the non-linearity problem. Zhang et al. [14] deployed random forests and gradient-boosted decision trees on financial ratio vectors, reporting accuracy gains over logistic baselines. These improvements are real — but they are *bounded* and *misleading*. Four structural problems afflict these approaches, each of which XGBoost cannot overcome regardless of depth or regularisation configuration:

Problem 1 — The Independence Ceiling. Both random forests and XGBoost operate on rows of a feature matrix where each row corresponds to one deal in isolation. The model learns per-deal statistical patterns but has no mechanism to propagate information across related deals — if two acquirers share a supplier and that supplier is distressed, both deals share latent risk that tabular models cannot detect. This is not a data engineering problem; it is an architectural impossibility.

Problem 2 — Survivorship Bias. M&A deal samples drawn from completed transactions systematically undersample value-destructive deals that were abandoned or never announced, biasing estimated coefficients toward the characteristics of deals that *proceeded* rather than deals that *succeeded* [13]. The consequence is that reported out-of-sample accuracy reflects the characteristics of deal survival, not value creation — a fundamental misdirection that no ensemble architecture can correct without restructuring the sampling frame.

Problem 3 — Look-Ahead Contamination. Feature engineering pipelines for M&A ML studies frequently construct ratios from annual filings that post-date the deal announcement,

embedding future information into the training set. Zhang et al. [14], for instance, discretise financial variables into categorical bins without explicit documentation of fiscal year alignment with announcement dates. This look-ahead contamination inflates validation accuracy beyond any economically realisable level.

Problem 4 — Feature Engineering as Crystallised Bias. Discretising continuous financial variables into categorical bins before tree construction (a common pipeline step in [14]) encodes human analyst heuristics into irreversible feature representations. The model learns from a discretised proxy for the raw signal — an automated version of the same heuristic reasoning employed by the human analysts whose systematic failures motivated building the model in the first place.

The combined effect of these four problems is an asymptotic accuracy ceiling: once the linear, rank-order, and non-parametric signal in tabular financial features has been captured, no further improvement is achievable through more sophisticated tabular models. This ceiling is *not* a function of computational power — it is a function of the information content of the feature space.

The Deep Learning False Dawn: MLPs and the Independence Assumption

The introduction of deep MLPs and LSTMs appeared to dissolve the linearity constraint. Elhoseny, Metawa et al. [15] achieved 95.8% accuracy on financial distress prediction using an Ameliorative Whale Optimization Algorithm-tuned deep neural network — a result that superficially suggests deep learning has resolved the prediction problem. It has not. The model processes each firm as a standalone entity, consuming individual financial metrics without any mechanism for inter-firm information exchange. Impressive accuracy on financial distress classification does not transfer to M&A synergy prediction: the former asks "is this firm failing?" from its own time series; the latter asks "will combining two firms create more value than they possess separately?" — a fundamentally relational question that non-relational architectures cannot answer.

The same architectural constraint applies to even the most sophisticated MLP implementations. A model with 10 layers and 10,000 parameters, trained on a vector of 50 financial ratios, still operates in a 50-dimensional space that is topologically flat: it contains no representation of whether the acquirer's top customer is about to go bankrupt, whether the target and acquirer share a fragile common supplier, or whether the deal creates a bottleneck in the industrial network. These omissions are not correctable by adding more layers.

Stream II gap: Every generation of tabular model — logistic regression, random forest, XGBoost, MLP, LSTM — inherits the independence assumption from its predecessor. The asymptotic ceiling is not a function of model complexity but of feature space topology. Breaking through it requires a graph operator (Stream IV). Before reaching that conclusion, Stream III examines why NLP approaches, despite initially appearing to transcend the tabular paradigm, ultimately reproduce the same structural failure.

Stream III — The Semantic Turn: Powerful Tools, Wrong Targets

The Bag-of-Words Era and the Loughran-McDonald Benchmark

The seminal contribution to financial NLP is Loughran and McDonald [16], who demonstrated that general-purpose sentiment lexicons systematically misclassify financial language: terms like "liability," "risk," and "obligation" carry negative sentiment in general English but are neutral

legal descriptors in financial filings. Their domain-specific word lists corrected this systematic mislabelling and established a critical methodological principle: financial text requires finance-specific analytical tools. This insight remains foundational and is directly inherited by this study's use of FinBERT over general BERT.

However, the bag-of-words (BoW) paradigm itself is architecturally insufficient beyond this correction. BoW models are "compositionally blind" [16]: they cannot distinguish "We have eliminated our exposure to risk" from "We have significant exposure to risk," because both sentences produce near-identical word frequency vectors while conveying directly opposite strategic signals. In the high-stakes context of M&A filings, where a single conditional clause ("subject to regulatory approval") can invert the meaning of a multi-billion dollar liability disclosure, this ambiguity is not marginal but catastrophic. Despite this, recent work continues to extend the BoW approach: Demers, Wang and Wu [17] construct frequency-based lexicons for human capital disclosures; Acheampong et al. [18] proxy financial constraints using lexicon-based indices; Garcia, Hu and Rohrer [19] develop finance dictionaries through market-based validation — all approaches that encode compositional blindness into increasingly refined vocabulary lists.

Hoberg and Phillips [20] represent the most structurally sophisticated BoW application: their text-based network industries classification (TNIC) used product-description cosine similarity from 10-K filings to construct time-varying firm similarity networks, demonstrating that textual proximity predicts competitive dynamics and deal *likelihood* with meaningful accuracy. This study directly inherits the semantic proximity methodology from Hoberg and Phillips [20] — but redirects it from deal occurrence to deal outcome, and replaces TF-IDF vectors with contextual FinBERT embeddings.

The Transformer Revolution and the Limits of Standard FinBERT

Devlin et al.'s [21] BERT architecture rendered static lexicon approaches architecturally obsolete by introducing contextual embeddings: dynamic vector representations where each word's encoding is a function of its surrounding context, enabling the model to resolve the polysemy that BoW approaches cannot. Araci's [22] FinBERT specialised this architecture through pre-training on a large corpus of corporate financial filings, giving it domain-specific representations that general BERT models lack.

However, even FinBERT — as deployed in existing M&A studies — falls into what this review terms the "Tabular Trap." Zhao, Li and Zheng [23] extracted BERT-based sentiment from M&A news articles and then *flattened the rich contextual output into a scalar sentiment score* before feeding it into a standard XGBoost classifier. This pipeline compresses a 768-dimensional semantic embedding — which encodes rich relational information about entities, relationships, and strategic context — into a single number, discarding the very representational richness that motivated using a transformer in the first place. The result is architecturally identical to an advanced BoW approach: the transformer acts as a sophisticated tokeniser, but its output is forced through the same bottleneck that limits lexicon-based methods.

Hajek et al. [24] and Han et al. [25] repeat this structural error in their M&A studies: both use transformer architectures to extract representations from financial text, but both predict binary deal *occurrence* (acquisition yes/no) rather than deal *outcome* (value-creating vs. value-destroying). This distinction is not incidental — it is the fundamental difference between a market microstructure problem (when will a deal be announced?) and a corporate strategy problem (will this deal create economic value?). Predicting deal occurrence requires detecting

announcement patterns in news flow; predicting deal outcome requires modelling the fundamental complementarity between acquirer and target across multiple information dimensions. The existing NLP literature has perfected the former and almost entirely ignored the latter.

Why Standard FinBERT Cannot Recover the Full Textual Signal

Beyond the "wrong target" problem, the standard FinBERT implementation used in the M&A literature has three additional structural limitations that this study directly addresses:

Limitation 1 — Frozen Generic Embeddings. Frozen FinBERT weights, pre-trained on a general corpus of financial filings, cannot adapt their representations to the specific semantic dimensions relevant to synergy prediction — particularly the distinction between *strategic alignment* (similarity in MD&A disclosures indicating compatible corporate strategies) and *risk concentration* (similarity in Risk Factor disclosures indicating overlapping vulnerabilities). This study exploits this distinction explicitly in H2, treating the two sections as semantically distinct modalities.

Limitation 2 — Single-Document Architecture. Studies that apply FinBERT to a single company's filing (acquirer or target independently) miss the *cross-document* signal: the pairwise semantic distance between acquirer and target disclosures. It is not the absolute strategic sophistication of either party that predicts synergy — it is their *relative alignment* across specific sections. This pairwise delta is what this study computes via cosine distance between paired acquirer-target embeddings.

Limitation 3 — Section Conflation. Standard implementations embed entire 10-K documents or undifferentiated excerpts. This conflates sections with opposite hypothesised relationships to CAR: MD&A similarity (strategic fit → positive CAR) and Risk Factor similarity (risk overlap → negative CAR). A model that conflates these sections is fitting a noisy mixture of opposing signals — a source of systematic downward bias on any textual feature's predictive coefficient.

The choice of FinBERT over Longformer [21] is justified on the following grounds: (i) domain-specific pre-training vocabulary alignment with the 10-K MD&A and Risk Factors sections; (ii) the section-specific extraction protocol used in this study obviates the long-document limitation of the 512-token context window; and (iii) Qin and Yang [26] demonstrated that frozen domain-specific representations with lightweight downstream heads yield competitive performance on financial classification tasks with sample sizes ($n < 5{,}000$) comparable to this study's deal universe.

Stream III gap: Transformer-based NLP architectures possess the semantic resolution to capture strategic fit signals in 10-K filings. They have not been applied to post-merger synergy outcome classification. Furthermore, even when correctly directed, textual models operating on isolated nodes miss the second-order structural reality: that two firms with perfectly aligned strategies can still fail to create synergy if their network positions make the combined entity more fragile. This structural incompleteness is what Stream IV addresses.

Stream IV — The Topological Turn: Network Structure as Irreducible Signal

The Economic Foundation: Supply Chain Momentum

The theoretical justification for encoding network topology into synergy prediction models is grounded in empirical financial economics, not computational novelty. Cohen and Frazzini [27]

established the foundational empirical result: economic shocks to a supplier do not immediately price into the customer firm's equity. Due to information friction and limited investor attention, these shocks propagate across supply chain links with a measurable time lag, generating a predictable momentum effect exploitable by investors who monitor inter-firm dependency structures. This "supply chain momentum" is direct empirical evidence that network topology encodes information about future firm value that is absent from standalone financial metrics — information that by definition cannot be captured by any model operating on single-firm feature vectors.

Ahern and Harford [28] extended this reasoning directly to M&A, demonstrating that the structure of industry-level trade networks predicts merger wave propagation: acquisitions cluster along supply chain linkages because buyers seek to internalise relationships that generate the highest dependency-reduction value. More critically for this study, Ahern and Harford [28] found that supply chain proximity at the industry level predicts post-merger combined stock returns — providing direct precedent that network position contains synergy-relevant information beyond financial fundamentals. This study operationalises this insight at the firm level using Bloomberg SPLC data, enabling node-level rather than industry-level topology encoding.

Corporate Network Centrality and Value

The broader corporate networks literature reinforces this foundation. Fee and Thomas [29] documented that customer-supplier relationships contain systematic pricing power information: firms with high customer concentration earn excess returns upon resolution of the relationship, confirming that vertical network dependencies have measurable equity value implications. Larcker, So, and Wang [30] demonstrated that director network centrality — betweenness centrality in board interlocks — is a significant predictor of firm future performance and acquisition premia, establishing the precedent that graph-theoretic centrality measures carry economic signal in corporate finance contexts.

The theoretical mechanism connecting these findings to H3 (Topological Arbitrage Hypothesis) is as follows: acquirers with high betweenness centrality in the supply-chain-competitor graph occupy bridge positions between otherwise disconnected industrial communities. These "bridge nodes" face bilateral dependency constraints — their centrality creates both upside synergy capture opportunities and downside exposure to disruption propagation. This bilateral constraint is hypothesised to compress the variance of |CAR| outcomes relative to peripheral nodes, as the diversifying effect of multiple network connections dampens both extreme positive and extreme negative market reactions.

Why XGBoost Cannot Recover the Topological Signal (Even With Graph Features)

A technically sophisticated critic might argue: cannot supply chain betweenness centrality simply be added as a hand-engineered feature to an XGBoost model? This approach — treating centrality as a scalar attribute — has two fundamental limitations that motivate a full GNN architecture.

First, scalar centrality measures collapse a high-dimensional structural signal into a single number, discarding the specific *pattern* of connectivity that carries economic meaning. Two firms can share identical betweenness centrality yet have radically different network contexts: one may bridge two healthy, high-growth industrial clusters; the other may bridge two sectors in secular decline. A scalar encoding treats these as identical; a graph embedding that propagates neighbourhood information does not. Hamilton, Ying, and Leskovec's [31] GraphSAGE

demonstrated exactly this: by learning aggregation functions over sampled neighbourhoods, the model recovers structural patterns that any scalar compression of the same graph would miss.

Second, scalar features cannot propagate *second-order* network effects. If an acquirer's primary supplier is itself a customer of the target, this creates a dependency loop that dramatically affects the risk profile of the combined entity — but this loop is invisible to any feature engineering pipeline that processes each pair independently. A message-passing GNN propagates information through multiple hop neighbourhoods by design, recovering these higher-order structural patterns without requiring their explicit enumeration.

Venuti [32] demonstrated GraphSAGE predicting acquisitions for private enterprise companies with 81.79% accuracy — concrete evidence of the inductive architecture's practical applicability to data-scarce M&A targets. Critically, Venuti's [32] accuracy gain over non-graph baselines was *largest* for private targets with sparse financial data — precisely the regime where graph neighbourhood information provides the largest marginal value over standalone features.

Heterogeneous Graphs: Why Edge-Type Semantics Matter

A homogeneous GNN that treats all inter-firm edges as equivalent commits a category error. A `supplies_to` relationship implies operational dependency and risk propagation: if the supplier fails, the customer's production line stops. A `competes_with` relationship implies market concentration and pricing power: if the two competitors merge, their combined market share may attract regulatory intervention. Collapsing these semantically distinct relationships into a single undirected edge type introduces a mixed signal that obscures what each relationship type independently encodes — analogous to averaging the coefficients of completely different explanatory variables.

Shi et al. [33] formalised the theoretical basis for Heterogeneous Information Networks (HINs) and demonstrated that type-specific attention mechanisms during neighbourhood aggregation consistently outperform homogeneous baselines on multi-relational graphs. Wang et al. [34] extended this into the Heterogeneous Graph Attention Network (HAN), showing that meta-path-based aggregation recovers semantic structure that single-type GCNs discard. Lv et al. [35] provided rigorous benchmarks confirming that heterogeneous graph encoders achieve statistically superior performance across multiple classification tasks on multi-relational corporate networks.

This study therefore constructs a heterogeneous graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{T}_v, \mathcal{T}_e)$ with semantically distinct edge encodings for `supplier_of`, `customer_of`, `competitor_of`, and `acquires` relationships within PyTorch Geometric's `HeteroData` structure. Type-specific GraphSAGE aggregation functions are applied independently per edge type before cross-type attention pooling — recovering exactly the semantic distinctions that scalar feature engineering and homogeneous GCNs discard.

Stream IV gap: GNN architectures are theoretically optimal for encoding inter-firm dependencies, and supply chain network structure is empirically demonstrated to contain synergy-relevant information. Yet no published study has directed a heterogeneous GNN at post-merger synergy outcome classification using firm-level topology. The Topological Alpha Hypothesis (H1), the Topological Arbitrage Hypothesis (H3), and the HeteroGraphSAGE architecture proposed in this study constitute the original contribution at this intersection.

The Measurement Foundation: Event Study Methodology

The Market Model and Its Implementation

The standard empirical tool for measuring M&A value creation is the event study, formalised by MacKinlay [36] and whose statistical properties were rigorously characterised by Brown and Warner [37]. The methodology rests on the market model: for each acquirer i, a normal returns generating process $R_{i,t} = \alpha_i + \beta_i R_{m,t} + \varepsilon_{i,t}$ is estimated over a pre-event estimation window, and Abnormal Returns (ARs) are computed as deviations from this expected return during the event window. The Cumulative Abnormal Return $\text{CAR}_i = \sum_{t=\tau_1}^{\tau_2} \text{AR}_{i,t}$ aggregates these deviations across the event window $[\tau_1, \tau_2]$.

MacKinlay [36] established that event window selection involves a fundamental bias-variance trade-off: narrow windows (e.g., $[-1, +1]$) minimise noise from confounding events but risk missing delayed market reactions for deals with information leakage; wider windows (e.g., $[-5, +5]$) capture more complete price discovery but increase variance. This study adopts $[-5, +5]$ trading days, consistent with the empirical M&A literature [13], as a balance that accommodates pre-announcement leakage (documented to begin 3–5 days before announcement in approximately 25% of transactions) while limiting contamination from post-announcement deal renegotiations.

Critical Assumptions and Their Limitations

Brown and Warner [37] identified three critical assumptions underlying the market model that warrant explicit scrutiny in M&A contexts. First, the *stationarity assumption* — that β_i is stable between the estimation window and the event window — is potentially violated for acquirers strategically repositioning ahead of the deal announcement. This study enforces an estimation window of $[-250, -10]$ with a 120-day minimum observation threshold, following MacKinlay's [36] recommendation, with deals below this threshold excluded from the final sample.

Second, the *benchmark contamination* risk: for mega-cap acquirers whose deal is large enough to shift the S&P 500 index, using the same index as the expected-returns benchmark creates a circularity. This study uses the value-weighted CRSP market index rather than the S&P 500 specifically to reduce single-stock concentration effects on the benchmark.

Third, Fama's [38] review of efficient markets evidence is particularly germane: the semi-strong form efficiency assumption embedded in event studies implies that announcement-date CARs represent unbiased estimates of the market's assessment of deal value. However, Fama [38] also documented systematic M&A anomalies — including long-run post-merger underperformance — inconsistent with strong-form efficiency. This study treats CAR as a noisy but informative signal of synergy potential, not a perfect measure, and uses binary directional classification rather than continuous regression to reduce sensitivity to measurement noise [13].

Justification for Binary Classification

The signal-to-noise characteristics of short-window CARs make continuous regression a challenging target for machine learning models. The pseudo-R^2 ceiling below 0.10 documented consistently across econometric studies [11], [13] is consistent with the Efficient Market Hypothesis's implication [38] that the *magnitude* of market surprise is largely unpredictable from pre-announcement information. The *direction* of CAR, however, contains learnable signal rooted in deal fundamentals — whether financial capacity is aligned with strategic intent and ecosystem health — that the feature space can realistically support. This study therefore defines $y_i = 1$ if $\text{CAR}_i > 0$ (value-creating) and $y_i = 0$ otherwise (value-destructive), with AUC-ROC

as the primary metric for its robustness to class imbalance and threshold-invariant evaluation of discriminative power [39].

The Multimodal Imperative: Why Fusion Is the Only Exit

Complementary Variance and the Failure of Mono-Modal Models

The four streams surveyed above converge on a single structural finding: financial, textual, and topological features encode *different*, partially non-overlapping aspects of synergy potential — what Baltrušaitis, Ahuja, and Morency [40] formalise as "complementary variance" in their taxonomy of multimodal learning architectures. A firm may exhibit strong capital adequacy (financial signal), precise strategic alignment (textual signal), yet suffer from fragile supplier dependencies (topological signal). A model observing only one dimension is mathematically blind to the divergence between the others, and no within-modality engineering can recover information that was never measured.

Xu et al. [41] demonstrated the practical validity of this argument in a financial forecasting context, showing that structured financial data and unstructured text features carry statistically independent predictive signal — signal that vanishes from each modality when the other is controlled for, but is recoverable when both are jointly modelled. Qin and Yang [26] similarly demonstrated complementarity between price-based and news-based features for stock movement prediction. Taken together, these results provide empirical support for the multimodal fusion architecture adopted in this study.

Why HeteroGraphSAGE Fusion Is the Only Logical Exit

Standard baselines — logistic regression (Stream II), XGBoost on ratios (Stream II), standalone FinBERT classifier (Stream III), homogeneous GCN with scalar centrality features (Stream IV) — each fail for a specific, architecturally irreversible reason. The table in Table 1 maps each failure mode precisely.

The HeteroGraphSAGE fusion architecture proposed in this study is designed to resolve each failure mode simultaneously: (i) GraphSAGE's inductive neighbourhood aggregation recovers the topological signal that tabular models cannot access; (ii) FinBERT's contextual embeddings on section-split 10-K filings recover the semantic signal that BoW and scalar NLP cannot represent; (iii) late fusion via concatenation preserves modality-specific representations while enabling cross-modal learning in the joint prediction head; and (iv) heterogeneous edge-type encoding preserves the semantic distinctions between `supplier_of`, `customer_of`, and `competitor_of` relationships that homogeneous GCNs discard.

This is not an incremental improvement over existing methods — it is a structural departure from the shared independence assumption that has constrained every prior generation of M&A predictive model. The following ablation design (Chapters 3 and 4) is specifically constructed to test whether this structural departure yields the statistically measurable accuracy gains that the theoretical argument predicts.

Architectural Choice: Late Fusion and Its Justification

The M&A deal universe with complete multimodal coverage is approximately 2,800–5,000 observations — far below the sample requirements for stable joint end-to-end training of transformer and GNN components [40]. This study therefore adopts *late fusion*: each modality is encoded independently into a fixed-dimensional embedding vector ($\boldsymbol{h}_F \in \mathbb{R}^d_F$, $\boldsymbol{h}_T \in \mathbb{R}^d_T$, $\boldsymbol{h}_G \in \mathbb{R}^d_G$), and these vectors are concatenated into a joint representation $\boldsymbol{z}_i = [\boldsymbol{h}_F \parallel \boldsymbol{h}_T \parallel \boldsymbol{h}_G]$ before a

shared MLP prediction head. This design isolates modality-specific representation learning from cross-modal inference, enabling robust individual stream training even when certain modalities have incomplete coverage — a practical necessity given that SPLC network data and EDGAR filings are not available for all deals in the LSEG sample.

The prediction head uses Chen and Guestrin's [42] XGBoost framework for the baseline ablation experiments, given its documented superiority over neural networks on heterogeneous tabular feature vectors at M&A sample sizes. The full multimodal MLP prediction head incorporates dropout regularisation [43] to control overfitting given the high feature-to-sample ratio. SHAP decomposition [44] is applied post-inference to provide per-modality attribution scores, enabling the controlled ablation experiments to quantify the marginal contribution of each block to predictive accuracy.

Synthesis: Gap Table and Research Hypotheses

The four knowledge streams converge on the synthesis table below, which maps each prior study to its structural failure mode and the specific architectural component of this study designed to overcome it.

Study / Stream	Method	Structural Failure Mode	This Study's Response
Palepu (1986); Barnes (1990)	Logit / MDA on ratios	Linearity bias; pseudo-$R^2 < 0.10$; no network features	Blocks A+B+C; non-linear fusion; graph topology
Zhang et al. (2024)	Random forest, XGBoost	i.i.d. assumption; survivorship bias; look-ahead contamination	Temporal train/val/test splits; GraphSAGE neighbourhood aggregation
Elhoseny et al. (2022)	Deep MLP (AWOA-DL)	High accuracy on distress; independence assumption unchanged; wrong target	Relational graph operator; synergy CAR as target
Zhao et al. (2020); Han et al. (2023)	BERT / RoBERTa → XGBoost	Tabular trap: embeddings flattened; predicts deal occurrence, not outcome	FinBERT on section-split 10-Ks; binary CAR classification (H2)
Hajek et al. (2024)	FinBERT sentiment	Single-document; no pairwise delta; acquisition likelihood target	Pairwise acquirer-target cosine distance; H2 conditional directionality
Loughran & McDonald (2011); Hoberg & Phillips (2016)	BoW / TF-IDF similarity	Compositional blindness; deal occurrence target	Contextual FinBERT embeddings; section-specific similarity (H2)
Cohen & Frazzini (2008); Ahern & Harford (2014)	Industry-level supply chain analysis	Industry-level only; no firm-level GNN; no CAR prediction	Firm-level GraphSAGE on SPLC data (H1, H3)
Venuti (2021)	Homogeneous GraphSAGE	Predicts deal likelihood; homogeneous edge types; no text or financial fusion	Heterogeneous edge types; full multimodal fusion
Alochukwu et al. (2024)	Dynamic GNN	Volatility prediction target; no M&A synergy outcome	CAR direction as prediction target; HeteroGraphSAGE architecture
Baltrušaitis et al. (2019); Xu et al. (2021)	Multimodal fusion frameworks	Not applied to M&A synergy; no graph modality	Full Block A+B+C late fusion with SHAP decomposition

Table 1: Synthesis of Four Literature Streams: Methods, Structural Failure Modes, and Architectural Responses

No prior published study has jointly fused financial fundamentals, FinBERT textual embeddings, and supply-chain graph topology, directed this multimodal architecture at binary CAR direction classification as a measure of post-merger synergy, and tested the marginal contri-

bution of each modality through controlled ablation. This trifecta of novelty constitutes the primary contribution of this dissertation.

Formal Research Hypotheses

The following three hypotheses emerge directly from the four-stream synthesis. Each is operationalised to map precisely to an ablation experiment in the empirical design.

H1 — The Topological Alpha Hypothesis: The inclusion of second-order neighbour embeddings via GraphSAGE (Block C) will yield a statistically significant increase in AUC-ROC ($p < 0.05$, paired Diebold-Mariano test) relative to the financial-only baseline (Block A), under 5-fold stratified cross-validation. This gain will be disproportionately concentrated within supply-chain-dependent manufacturing sectors (SIC 20–49) compared to asset-light service sectors (SIC 60–79), reflecting the hypothesis that graph embeddings recover signal proportional to industrial ecosystem structural density [27], [28].

H2 — The Semantic Divergence Hypothesis: The predictive relationship between textual similarity and synergy direction is *conditional* on document section. High cosine similarity in strategic disclosures (MD&A) will positively correlate with CAR — capturing strategic fit — whereas high similarity in Risk Factors will negatively correlate with CAR — capturing risk concentration. This conditional directionality is tested via bivariate OLS $\text{CAR}_i = \beta_0 + \beta_1 \cdot \text{sim}_{\text{MDA},i} + \beta_2 \cdot \text{sim}_{\text{RF},i} + \varepsilon_i$, and directly refutes the monotonic sentiment utility assumption embedded in [16] and [24].

H3 — The Topological Arbitrage Hypothesis: Acquirer nodes exhibiting high betweenness centrality in the heterogeneous supply-chain graph will exhibit statistically compressed variance in |CAR| outcomes relative to peripheral nodes, as measured by Levene's test across betweenness centrality quantile groups [28], [30]. The full multimodal fusion model (Blocks A+B+C) will achieve statistically superior AUC-ROC relative to all single-modality baselines, confirming the multimodal complementarity established by [40] and [41].

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